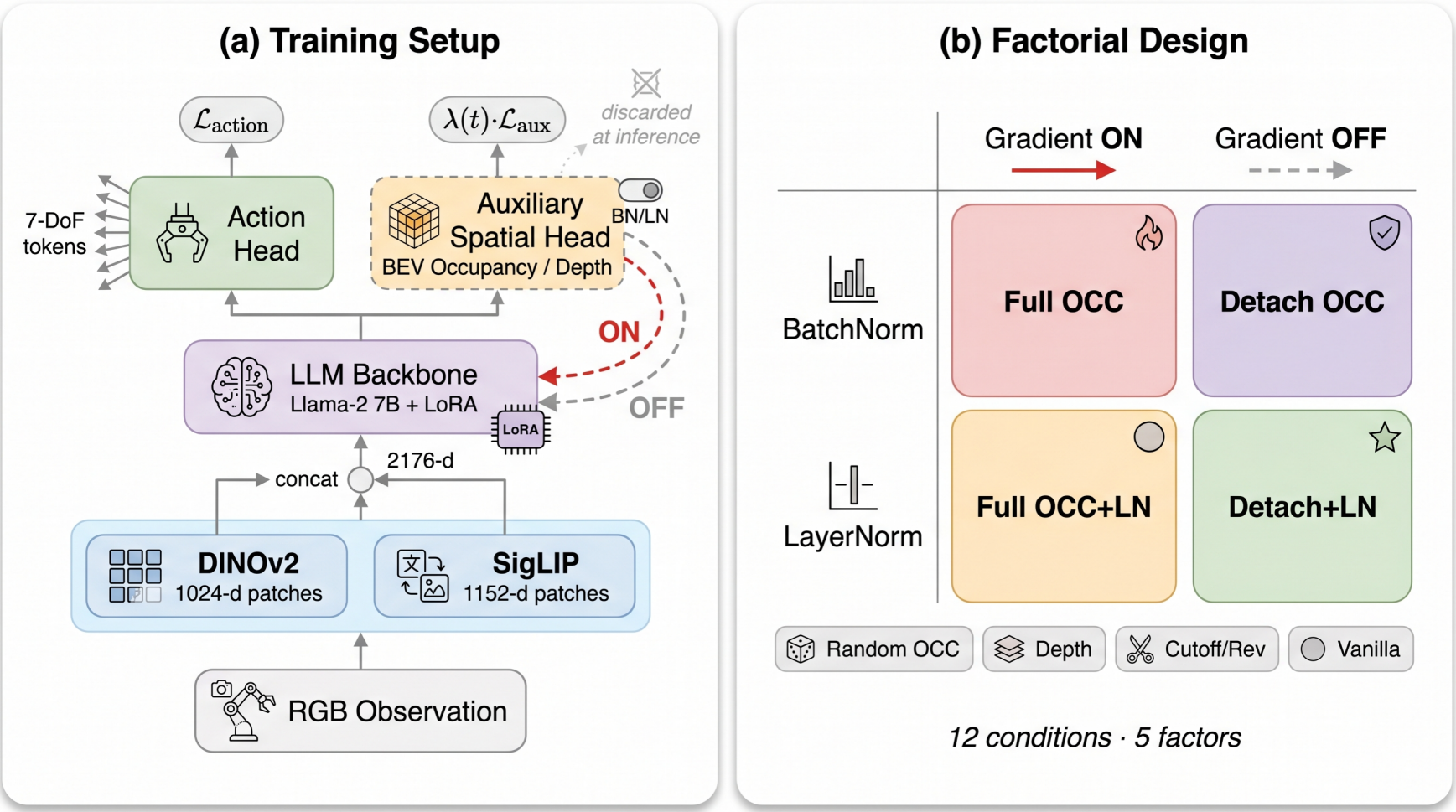

(a) Training Setup
$\mathcal{L}_{\text{action}}$
$\lambda(t)\cdot\mathcal{L}_{\text{aux}}$
discarded at inference
7-DoF tokens
Action Head
Auxiliary Spatial Head
BEV Occupancy / Depth
BN/LN
ON
OFF
LLM Backbone
Llama-2 7B + LoRA
LoRA
2176-d
concat
DINOv2
1024-d patches
SigLIP
1152-d patches
RGB Observation
(b) Factorial Design
Gradient ON
Gradient OFF
BatchNorm
LayerNorm
Full OCC
Detach OCC
Full OCC+LN
Detach+LN
Random OCC
Depth
Cutoff/Rev
Vanilla
12 conditions · 5 factors